Module 2 - Project Selection and Methodology

We can’t do ALL the projects!

- The projects should be selected strategically → It should contribute to business goals
- **In Industry**
  - **Project Portfolio Management (PPM)** : A process for selecting, prioritizing and monitoring project results.
  - There are **tools** that help with PPM.

Project Management

- Once a project is selected, it should be **managed.**
- Managing is about **time**, **cost** and **desired outcome.**
- There are certain issues in managing a project that matter to a systems analyst:
  - A **methodology** that is **appropriate** for a given project should be selected.
  - **Staffing** needs are determined
  - **Coordination** methods are established
  - Project **risks** are mentioned and managed.

Lesson 1 - Project Selection

How do we select a project? What factors do we consider when making a decision?

PLANNING

TASK CHECKLIST

- [x] Identify project.
- [x] Develop systems request.
- [x] Analyze technical feasibility.
- [x] Analyze economic feasibility.
- [x] Analyze organizational feasibility.
- [ ] Perform project selection review.
- [ ] Estimate project time.
- [ ] Identify project tasks.
- [ ] Create work breakdown structure.
- [ ] Create PERT Charts.
- [ ] Create Gantt Charts.
- [ ] Manage scope.
- [ ] Staff project.
- [ ] Create project charter.
- [ ] Set up CASE repository.
- [ ] Develop standards.
- [ ] Begin documentation.
- [ ] Assess and manage risk.

Project Selection

- **Several** important initiatives simultaneously
- A **committee** that selects
- Projects are not considered in **isolation**
- Companies:
  a. **prioritize** their business **strategies**
  b. Assemble and assess project portfolios (**a collection of projects**)

Project Portfolio

- Projects could be large or small, high risk or low risk, strategic or tactical.
- A good project portfolio will have **the most appropriate mix of projects**.
- Beyond just cost and benefits, the **technical and organizational risks** should be considered as well.

Size	What is the size? How many people are needed to work on the project?
Cost	How much will the project cost the organization?
Purpose	What is the purpose of the project? Is it meant to improve the technical infrastructure? Support a current business strategy? Improve operations? Demonstrate a new innovation?
Length	How long will the project take before completion? How much time will go by before value is delivered to the business?
Risk	How likely is it that the project will succeed or fail?
Scope	How much of the organization is affected by the system? A department? A division? The entire corporation?
Economic Value	How much money does the organization expect to receive in return for the amount the project costs?

FIGURE 2-1 Ways to classify projects.

❏ Reading - Project Selection

Lesson 2 - Various Methodologies

What is a methodology?

- Remember that SDLC has four fundamental phases and that each phase includes tasks, steps and deliverables. These components can come together in various **patterns**.
- A **methodology** is a formalized approach to **implementing the SDLC**
- There are **predominant methodologies** that have evolved over time.
- An organization can develop their **own methodology**.

There is no *best* methodology

- Depending on the characteristics of each project, one method might be better than another:
  - Clarity of user requirements
  - Familiarity with technology
  - System complexity
  - System reliability
  - Short time schedules
  - Schedule visibility

Waterfall Development

- Proceeds **sequentially** from one phase to the next
- If a phase is **approved**, it ends and the nest phase begins
- Quite difficult to go **backwards**
- Deliverables for each phase are **voluminous**!

- **Advantage:** Requirements do not change.

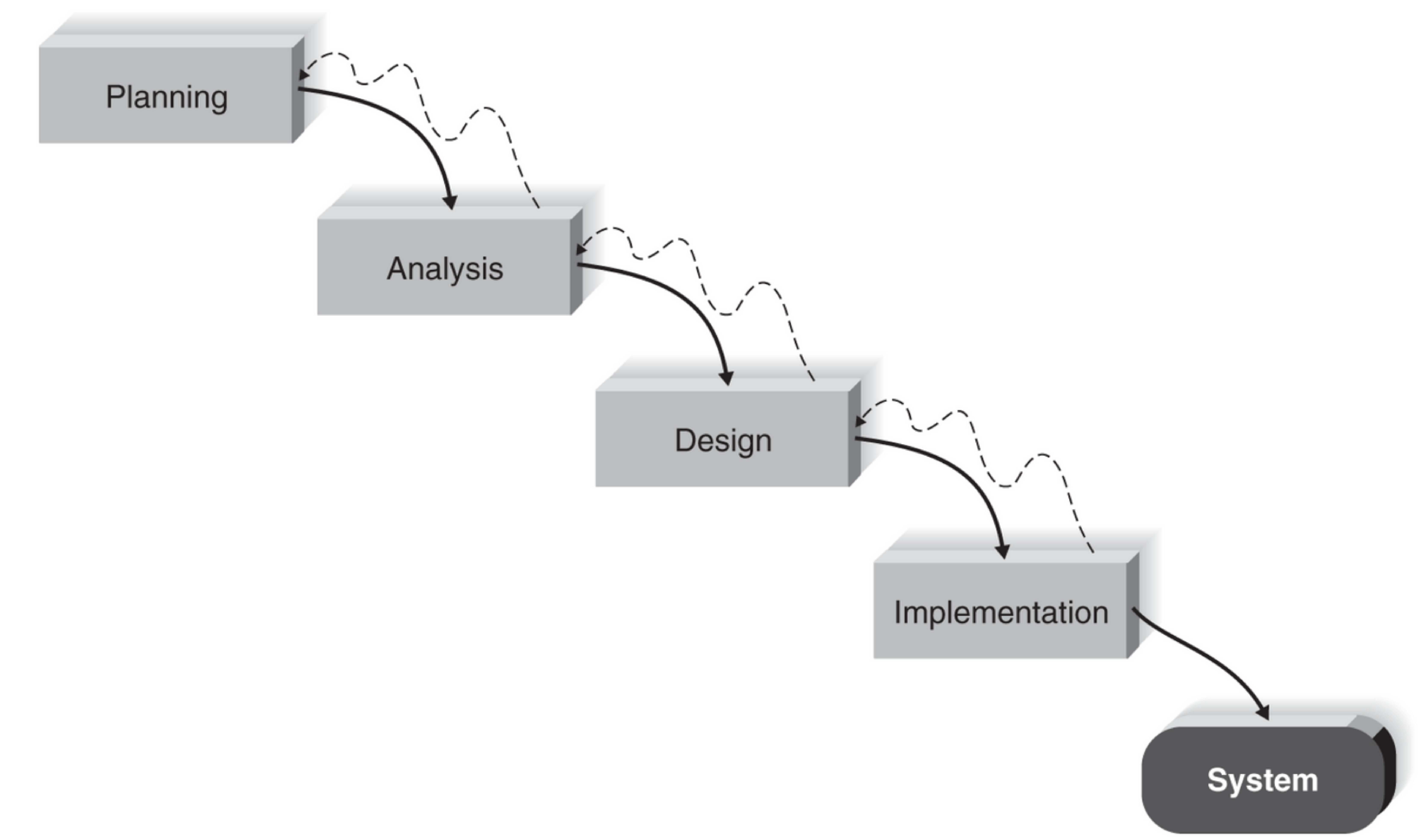

FIGURE 2-2 **Waterfall development.**

Disadvantages of Waterfall Development

- **Design must be completely specified before programming begins.**
- A big **time gap** between system proposal and delivery
    - Environments change so quickly that what we wanted back then might no longer be relevant! And users might forget the purpose of the system.
- **Testing** is often treated as an afterthought.
- Deliverables are not the best form of **communication**
    - Important requirements might be overlooked in the volumes of documentation.

There are **variants of waterfall** developed to address some of these issues.

Parallel Development

- To address the **time gap**.
- A **general design** is created. (the idea is *shaped* enough)
- **Sub-projects** then would do more in-depth design (parallel teams)

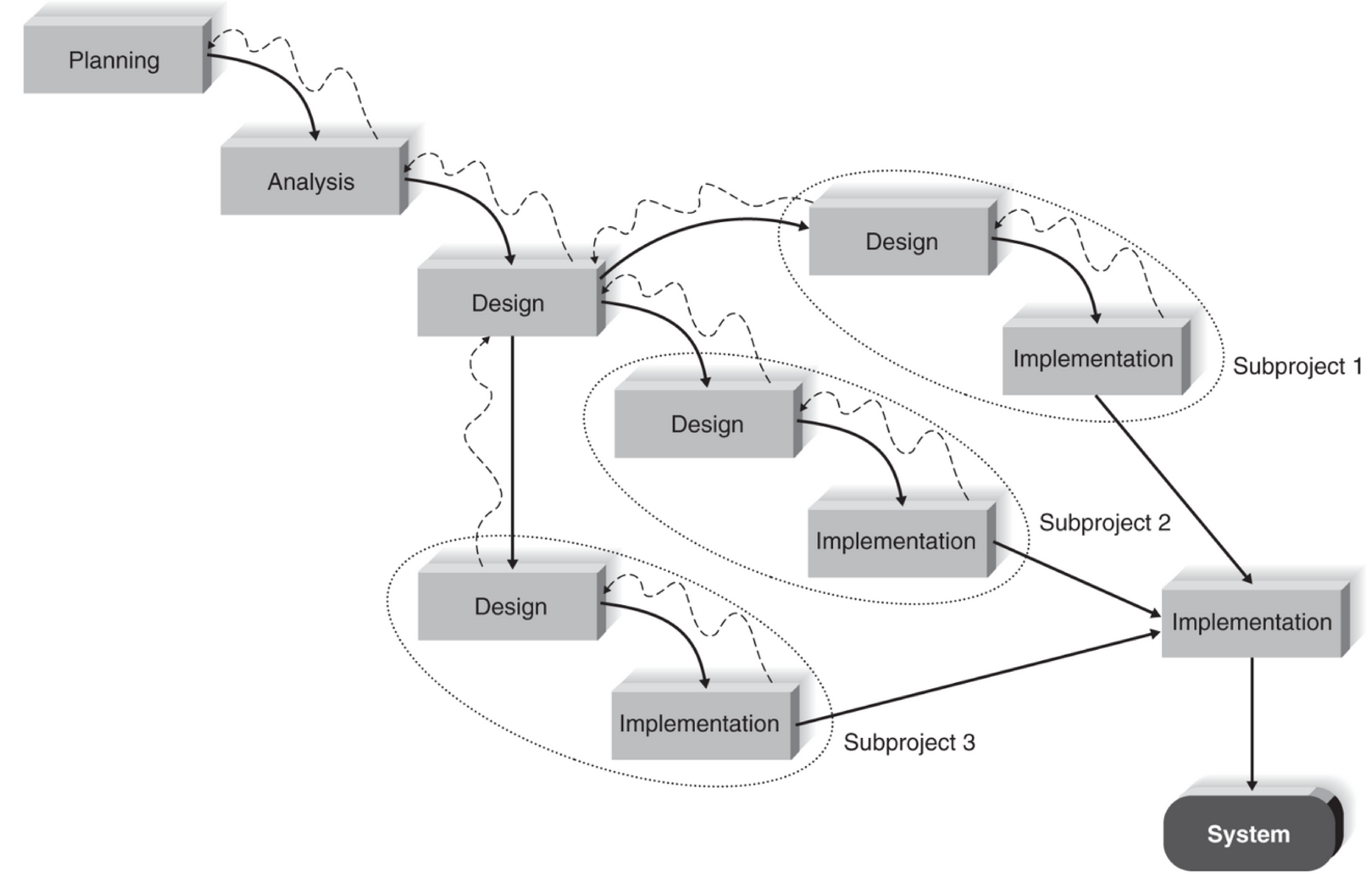

FIGURE 2-3 **Parallel development.**

V-Model

- Specific attention to **testing**.
- As requirements are specified and components are designed, testing for those elements is also defined.

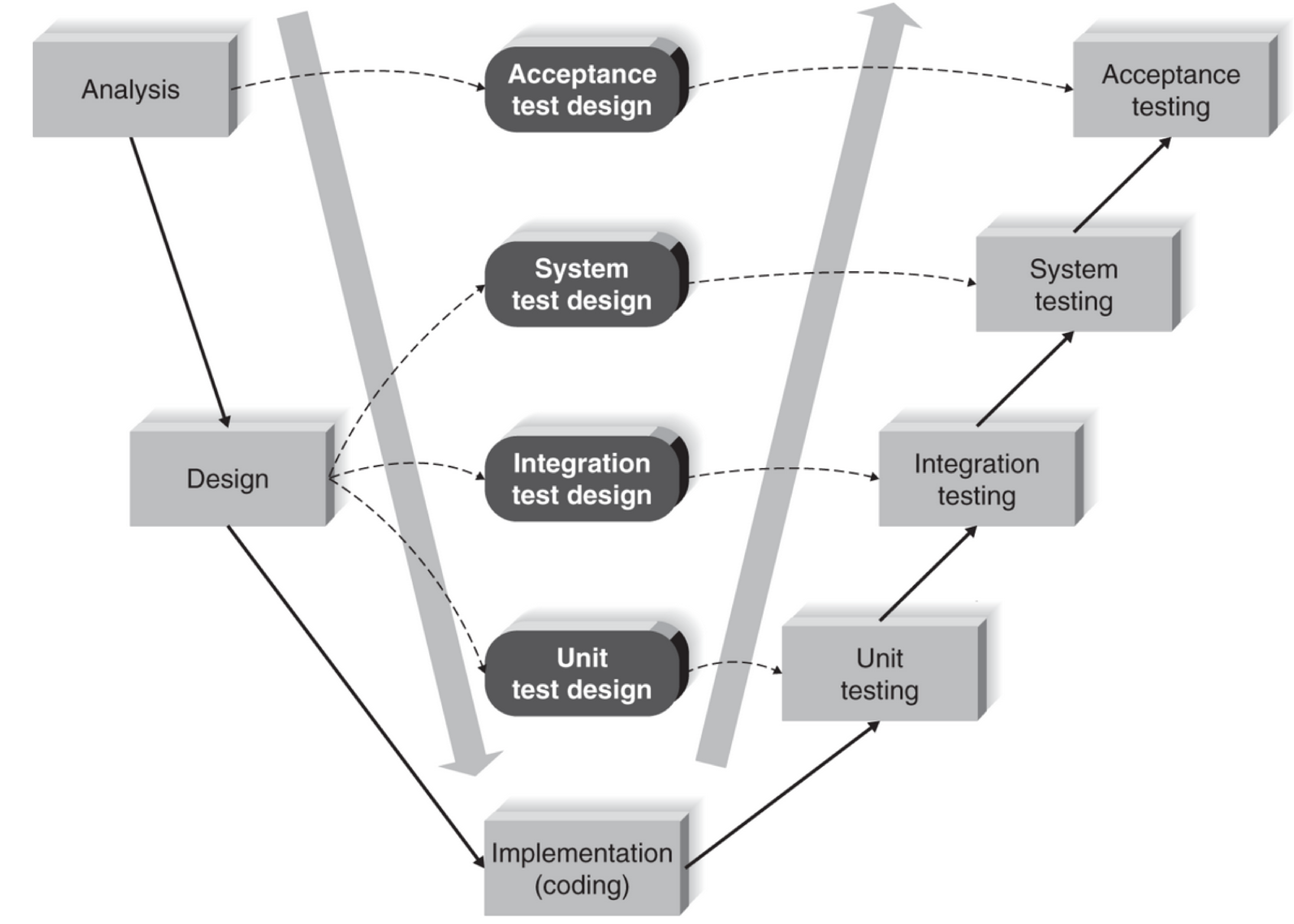

FIGURE 2-4 V-Model.

Rapid Application Development

- **A collection** of methodologies
- Emerged **in response to the weaknesses of waterfall** and its variations.
- It tries to speed up the analysis, design and implementation to get some portion of the system into the hands of users for **evaluation and feedback**.
- Conducted in variety of ways:
  - Iterative development
  - System prototyping
  - Throwaway prototyping

Iterative Development

A **series of versions** that are developed sequentially.

Most appropriate when:

- **timelines** are short
- when project **visibility** is particularly important.

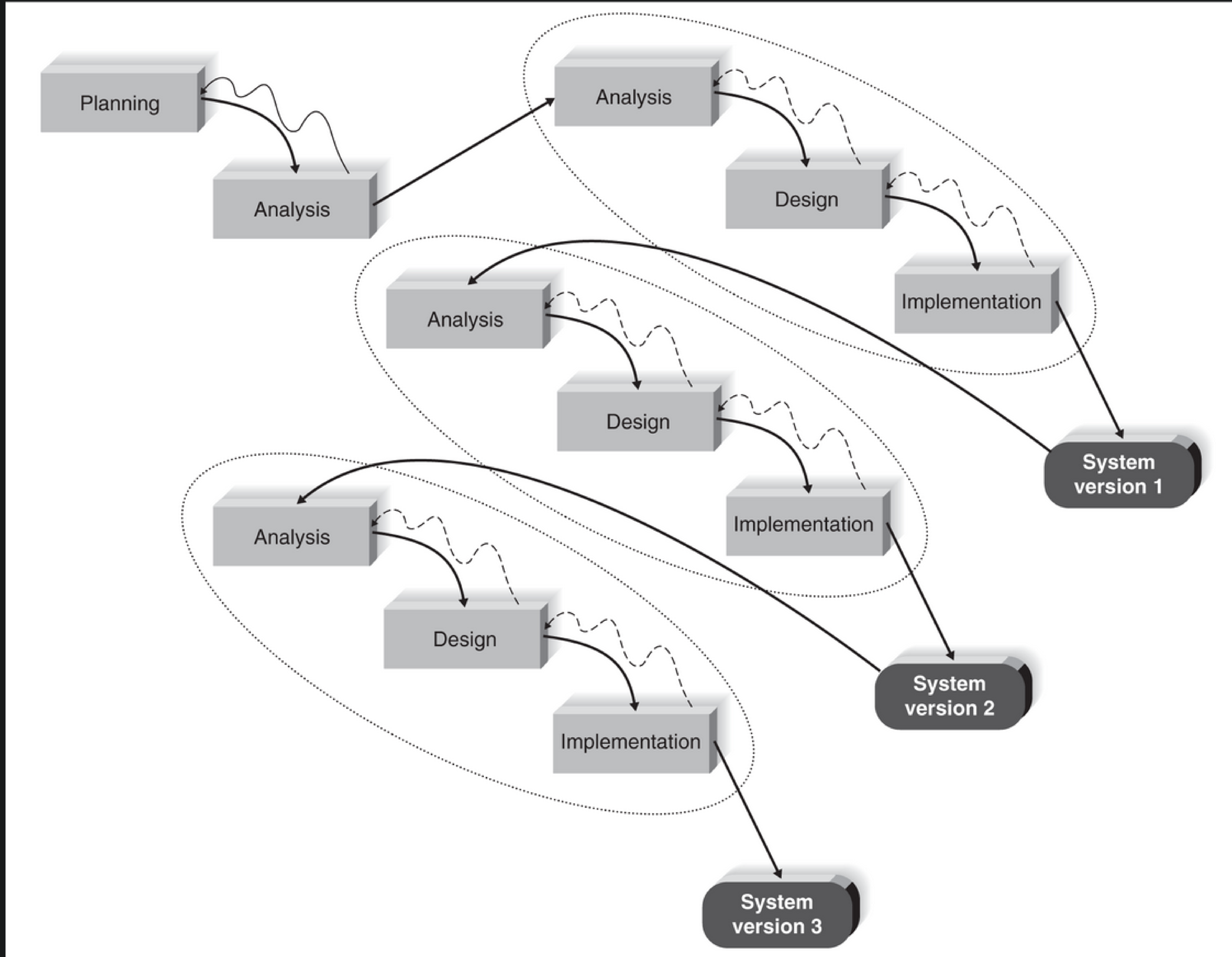

FIGURE 2-5 Iterative development.

System Prototyping

Performs analysis, design, and implementation **concurrently** to give users a “**quick and dirty**” version for evaluation and feedback. The prototype then becomes the actual new system.

Most appropriate when:

- user **requirements** are unclear
- **timelines** are short
- when project **visibility** is particularly important.

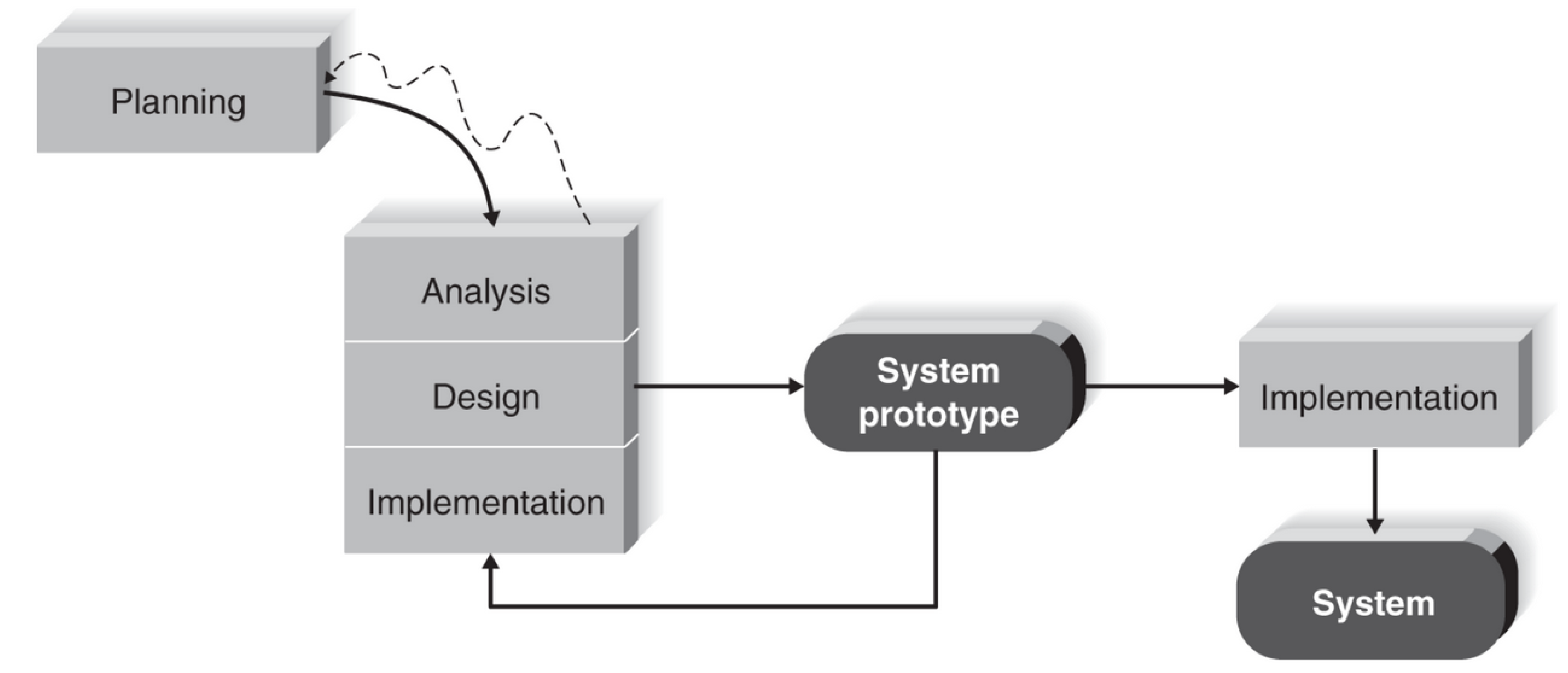

FIGURE 2-6 **System prototyping.**

Throwaway Prototyping

Develops prototypes but only to explore **design alternatives**.

Most appropriate when:

- user **requirements** are unclear
- the system will use new, **unfamiliar technology**
- the system is **complex**
- system **reliability** is a high priority

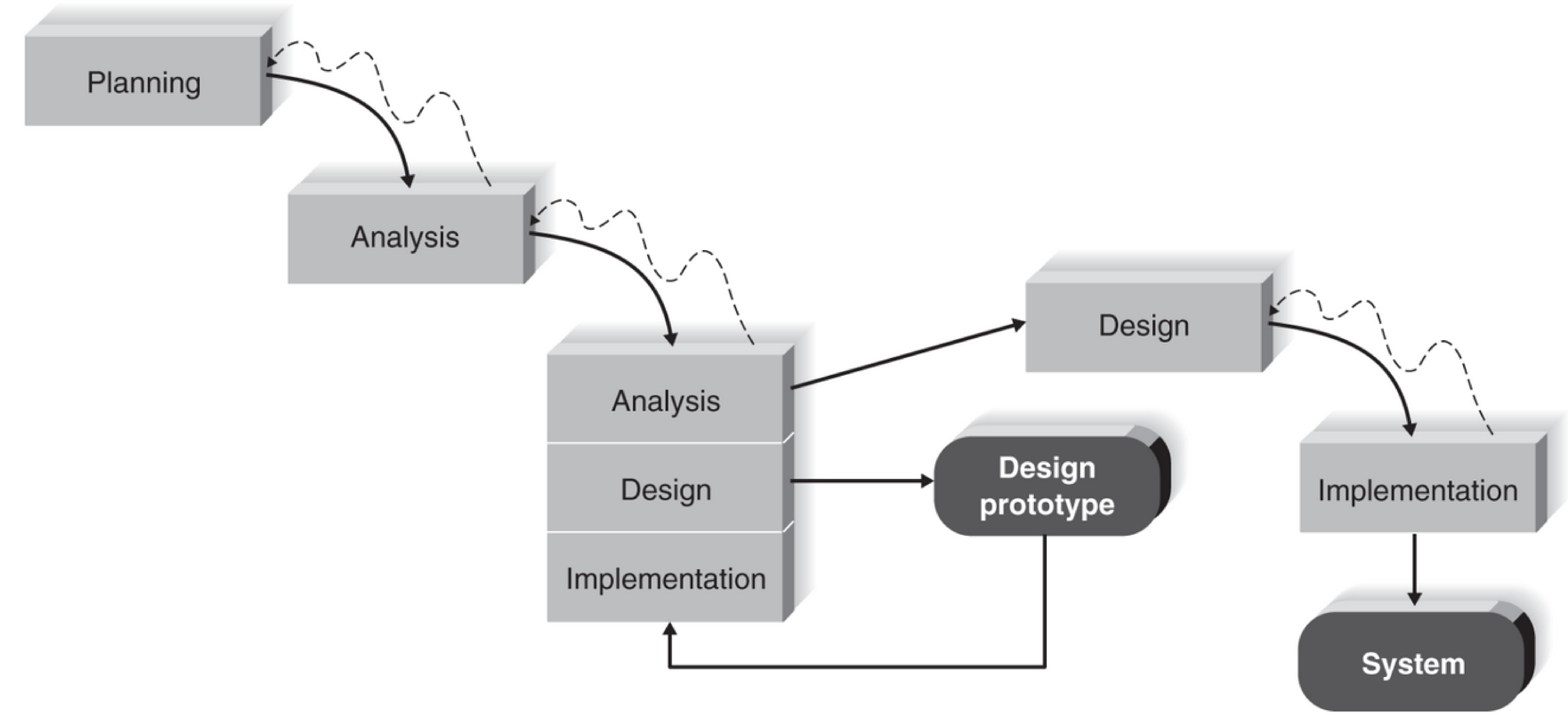

FIGURE 2-7 Throwaway prototyping.

❏ Research: Methodologies Used in Industry

Agile Development Methods

- **Business Agility:** Being able to respond quickly and effectively to a dynamic business environment and customer demand.
    - Traditional methodologies are often deemed too **rigid**, **slow** and **inflexible**.
- Agile development is a group of **programming-centric** methodologies that focus on streamlining the SDLC:
    - Modeling and documentation overhead is eliminated
    - Face-to-face communication is preferred
    - Every iteration is a complete project
    - Short cycles (1- 4 weeks)

Agile Values

"While there is value in the things on the right, we value the things on the left more."

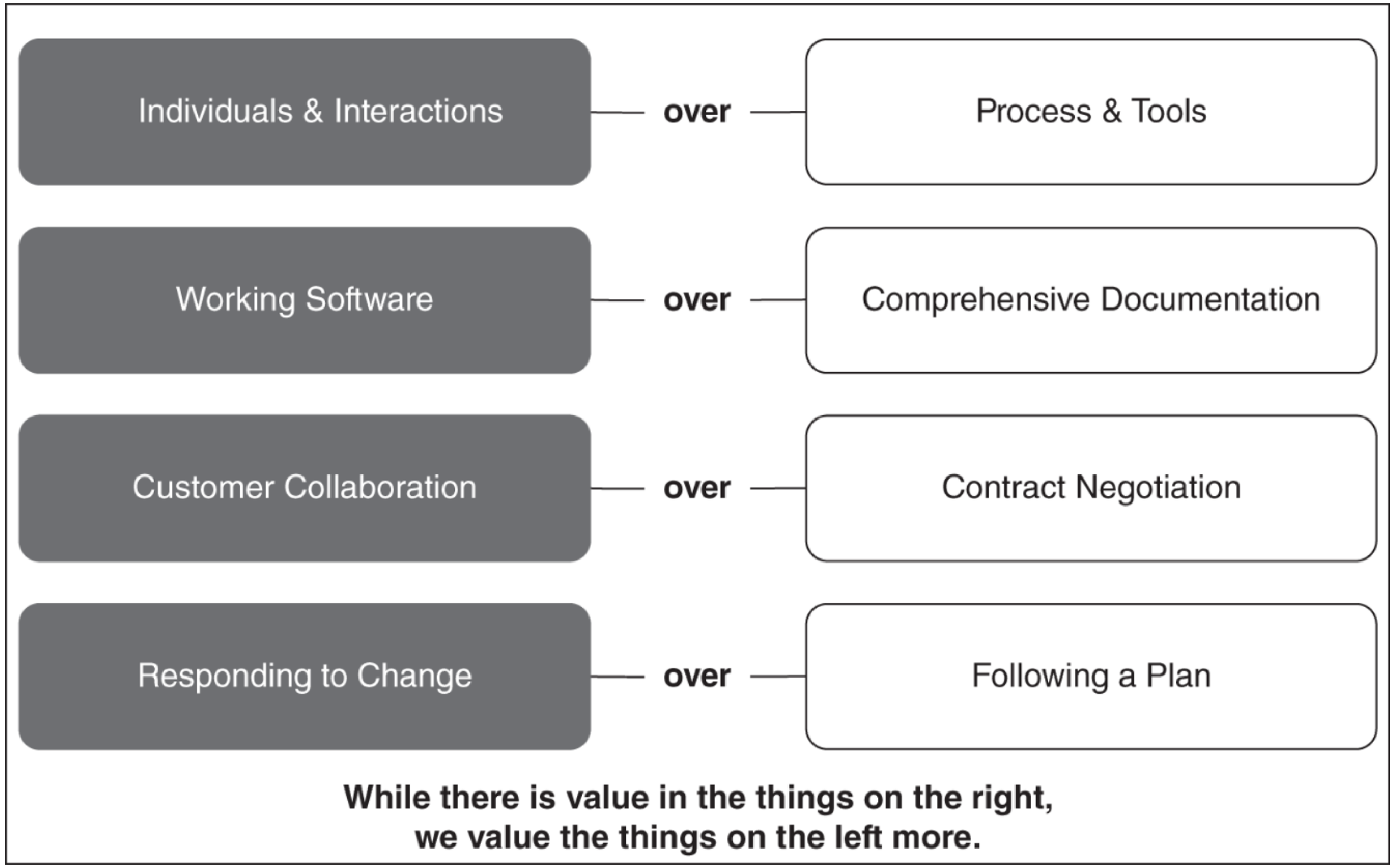

FIGURE 13-1 Agile Manifesto values.

Agile Principles

1. Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.
2. Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.
3. Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.
4. Businesspeople and developers must work together daily throughout the project.
5. Build projects around motivated individuals. Give them the environment and support they need and trust them to get the job done.
6. The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.
7. Working software is the primary measure of progress.
8. Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.
9. Continuous attention to technical excellence and good design enhances agility.
10. Simplicity—the art of maximizing the amount of work not done—is essential.
11. The best architectures, requirements, and designs emerge from self-organizing teams.
12. At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

FIGURE 13-2 **Agile Manifesto principles.**

❏ Reading - Agile Principles

Summary of Agile Principles

- Close **collaboration** between the project team and business experts
- Face-to-face **communication**
- Frequent **delivery** of new, deployable business value
- Tight, **self-organizing** teams
- **Reduced impact** of changes in requirements

These principles are the foundation for **all agile methodologies**.

Popular agile approaches

- **Scrum (the most common)**
- Extreme programming (XP)
- Dynamic Systems Development Method (DSDM)

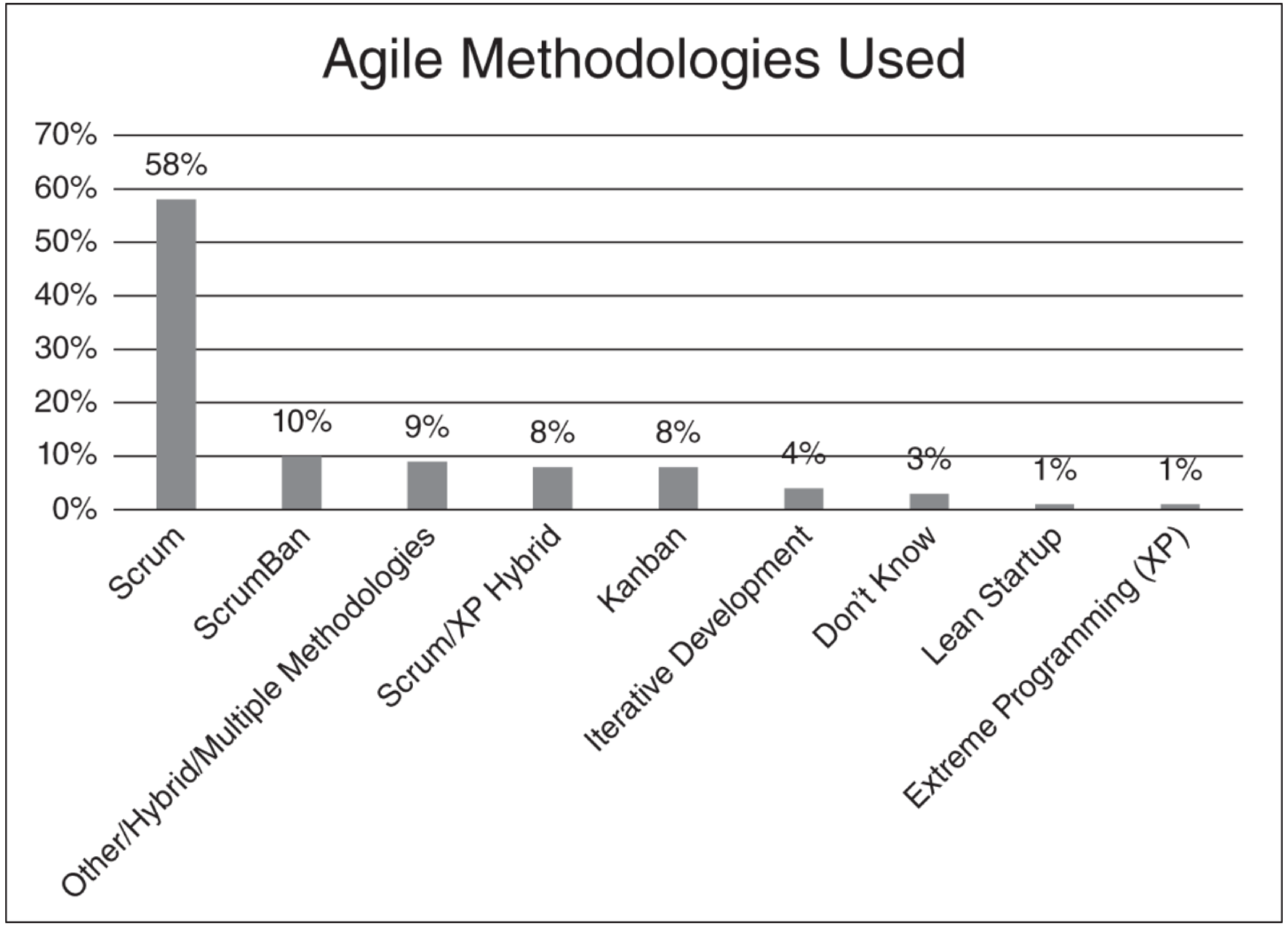

FIGURE 13-4 Agile methods in use.

Organizations might **mix and match** aspects of different agile methodologies.

Scrum

- 2-4 week **cycles**
- Business **priorities** define the “to-do” list
- The team **self-organizes** to determine the best way to deliver the features in response to these priorities
- Every 2-4 weeks, the team demonstrates **real working software**.
- Versatile approach, used in:
  - Large and small organizations
  - Array of industries
  - Wide range of applications

Overview of Scrum

- **Product backlog** represents the system requirements
- Product backlog serves as the team's **to-do list**
- **Product owner** manages the priorities of the features listed in the backlog
- **Cycles** of work are called sprints

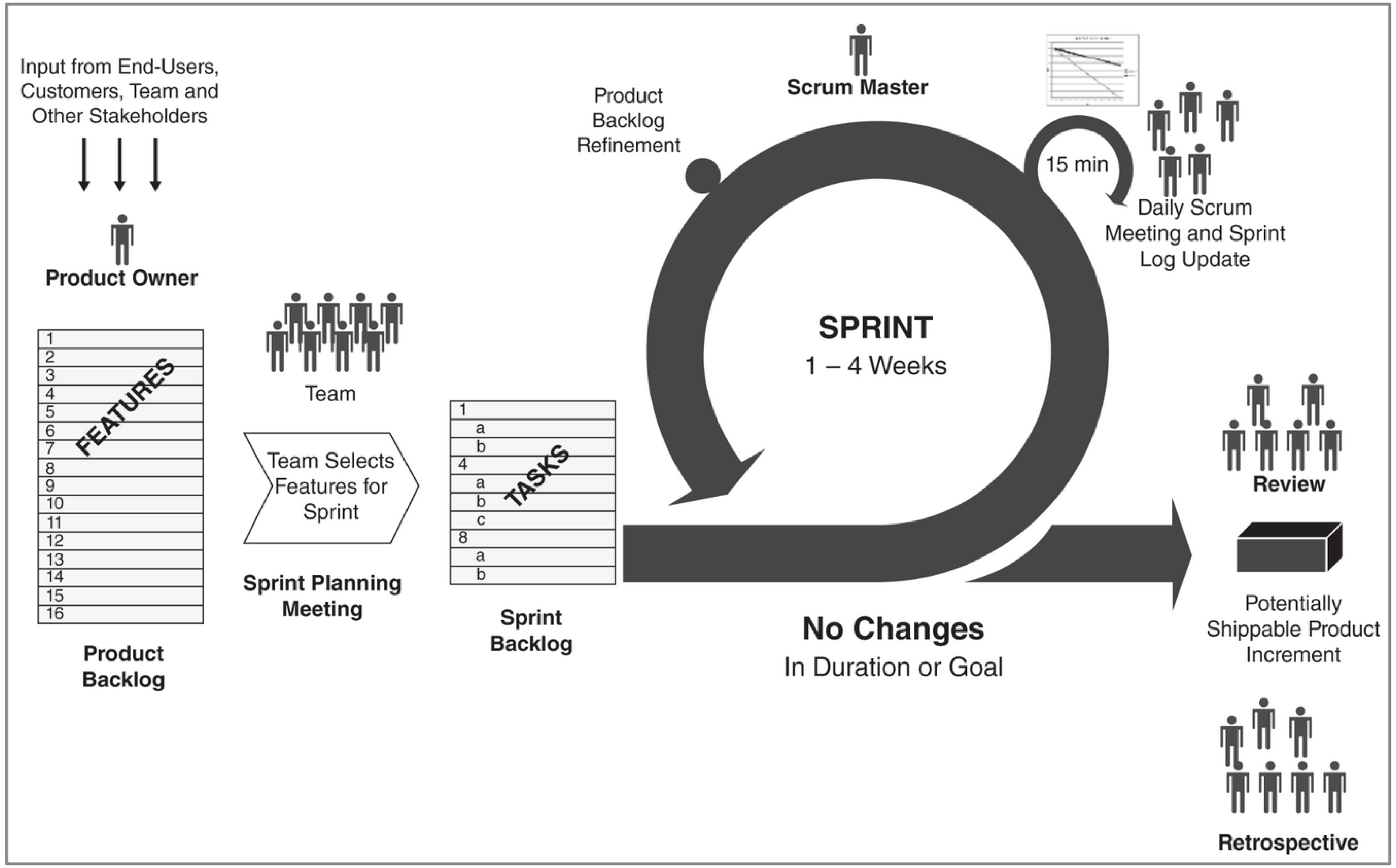

FIGURE 13-5 Scrum overview.

Sprints

- 2-4 weeks. That’s why they are called sprints!
- At the beginning of each sprint:
    - Review the backlog
    - Decide what to work on
    - Refine selected features into a more detailed set of tasks → **Sprint Backlog**
- During the Spring
    - Design, code and test software
    - Meet daily and report on your progress (**daily scrum**)

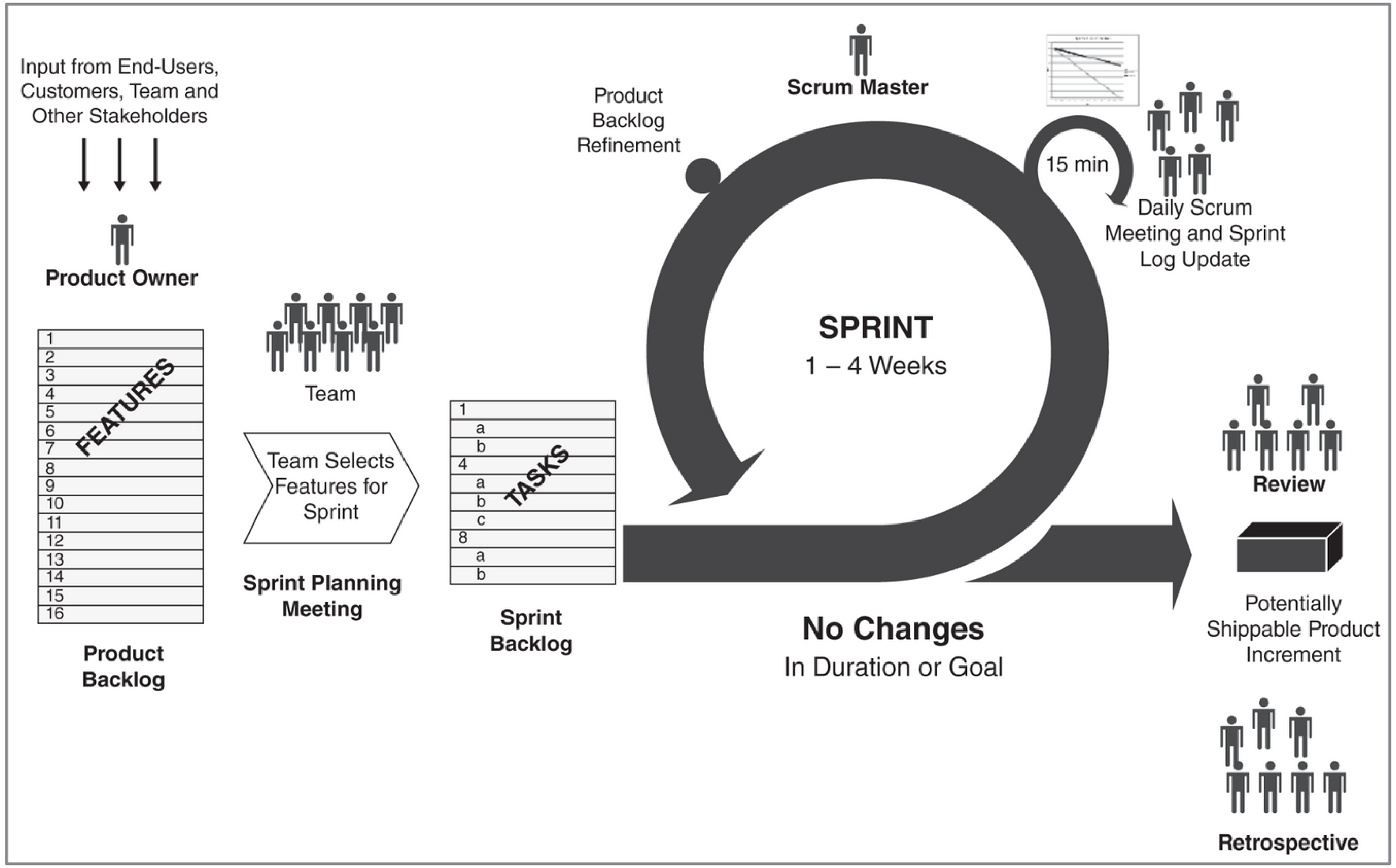

FIGURE 13-5 Scrum overview.

End of Cycle

- **Demonstrate** what was done
- Can the completed work be **integrated** into the product? Or does it require more work in the next sprint?
- The scrummaster and product owner will refine the prioritized product backlog to reflect what is done, and the **new understandings** that have emerged.

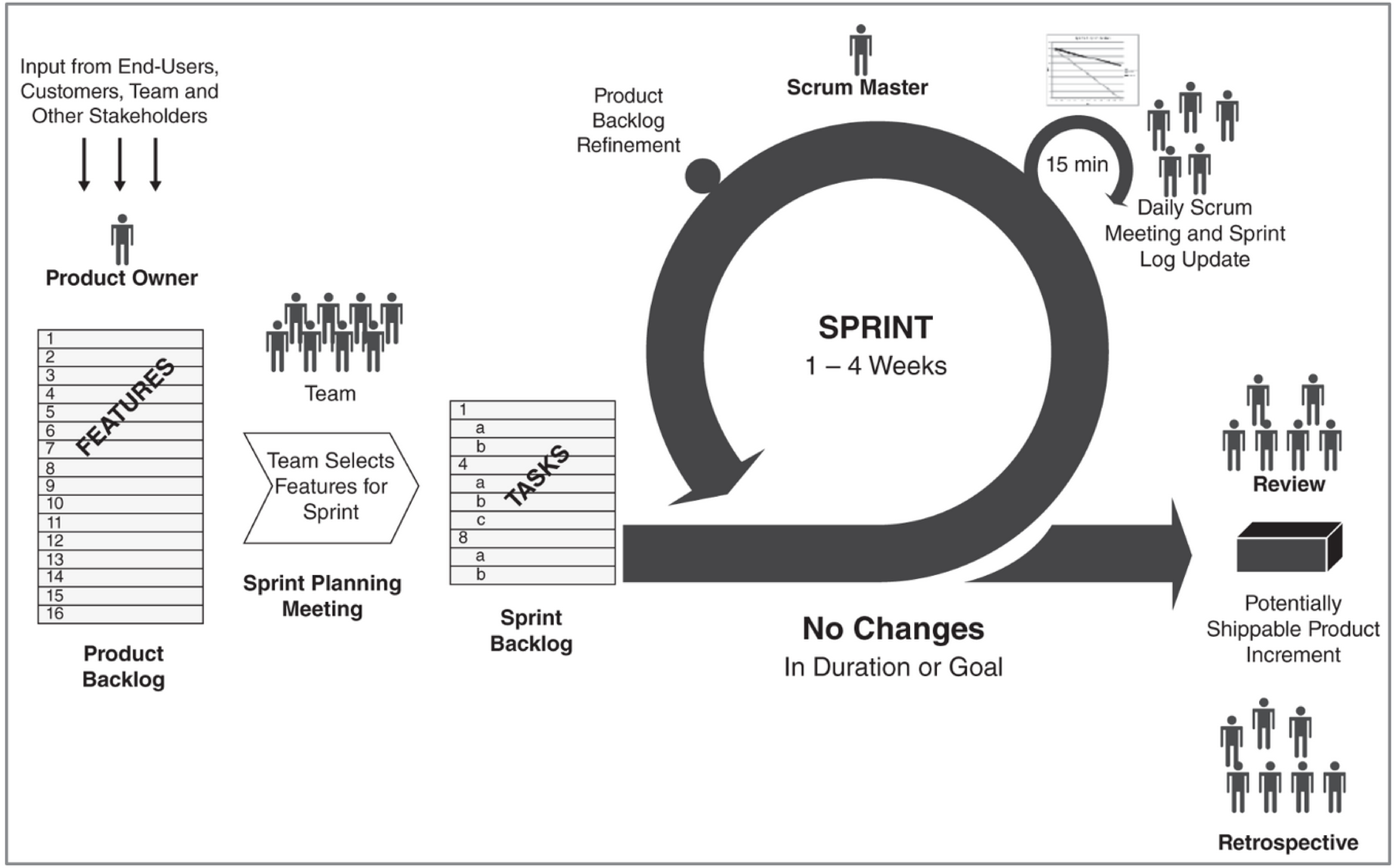

FIGURE 13-5 Scrum overview.

Your Turn: Other Cyclic Approaches

- Reading - Six-week cycles

Lesson 3 - Selecting a methodology

Agile vs. Waterfall

- Agile approaches were created partly because of dissatisfaction with the sequential, inflexible structure of waterfall-based approaches.
- Suggesting that an organization would be “all-agile” or “all-waterfall” is a false choice.
- **Many software developers actively seek to integrate best elements of both waterfall and agile into their practice.**
- Hybrid agile-waterfall approaches are evolving.
- The process is never static.
- A best methodology depends on what you are trying to do. (the project)

Selecting the appropriate methodology

- Many organizations have **standards and policies** to guide the choice of methodology.
  - an “approved” methodology
  - Several methodology options
  - Or no formal policies
- Each methodology has **strength and weaknesses**

Criteria for selecting a methodology

Waterfall

RAD

Usefulness in Developing Systems	Waterfall	Parallel	V-Model	Iterative	System Prototyping	Throwaway Prototyping	Agile Development
With unclear user requirements	Poor	Poor	Poor	Good	Excellent	Excellent	Excellent
With unfamiliar technology	Poor	Poor	Poor	Good	Poor	Excellent	Poor
That are complex	Good	Good	Good	Good	Poor	Excellent	Poor
That are reliable	Good	Good	Excellent	Good	Poor	Excellent	Good
With short time schedule	Poor	Good	Poor	Excellent	Excellent	Good	Excellent
With schedule visibility	Poor	Poor	Poor	Excellent	Excellent	Good	Good

FIGURE 2-9 **Criteria for selecting a methodology.**